



Semantic Segmentation for Autonomous Vehicles

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Introduction

Autonomous vehicles must understand their surroundings in real time to navigate safely. One critical component of this is semantic segmentation, or assigning a class label to every point in a scene. While camera-based segmentation has been extensively studied, LiDAR-based segmentation is able to provide direct 3D geometric information and is robust to lighting conditions that challenge RGB cameras.

Using the SemanticKITTI dataset, a large-scale urban driving benchmark, we tackle LiDAR-based segmentation. Rather than operating on unstructured 3D point clouds, we project each LiDAR scan into a 2D range image, enabling the use of standard CNNs. We train RangeNet++, a CNN that classifies each pixel of the range image through an encoder-decoder architecture.

Our work examines how well a CNN trained on range images generalizes across a driving sequence, analyzing per-class performance to see where the model succeeds and where class imbalance or geometric ambiguity poses challenges.

Preprocessing

Range Image Embedding

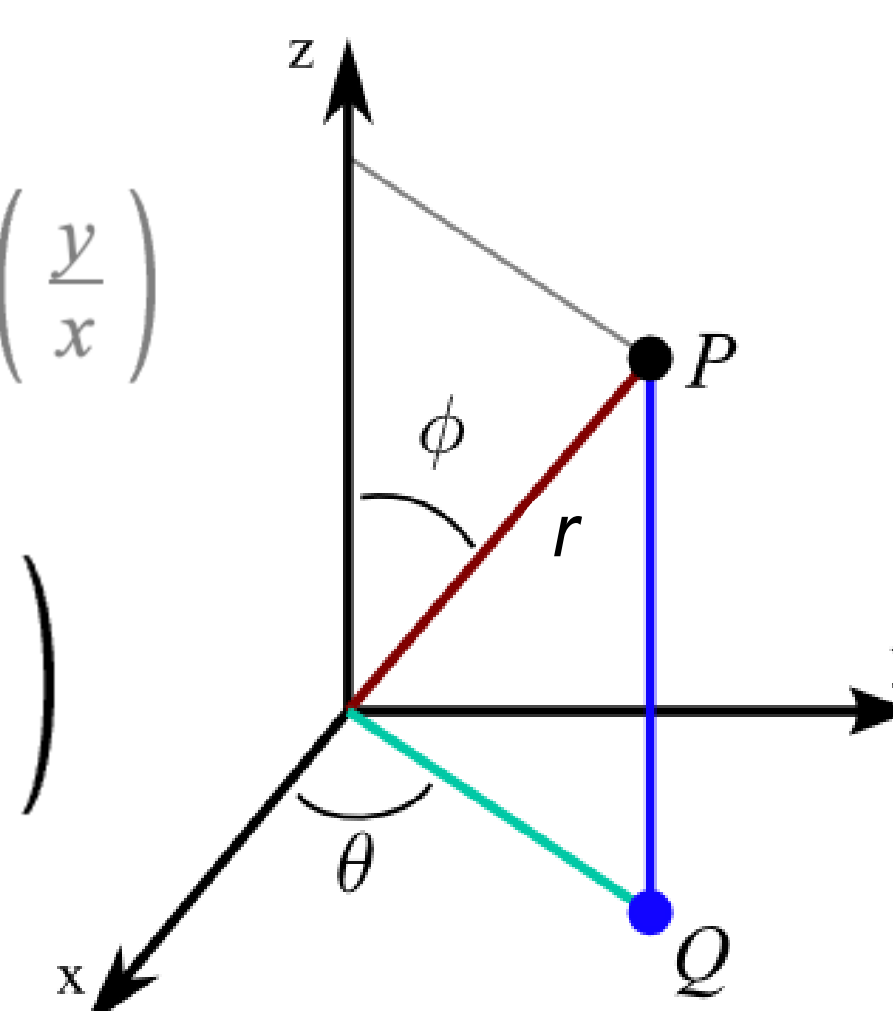
f : field of view

$h=64$: range image height

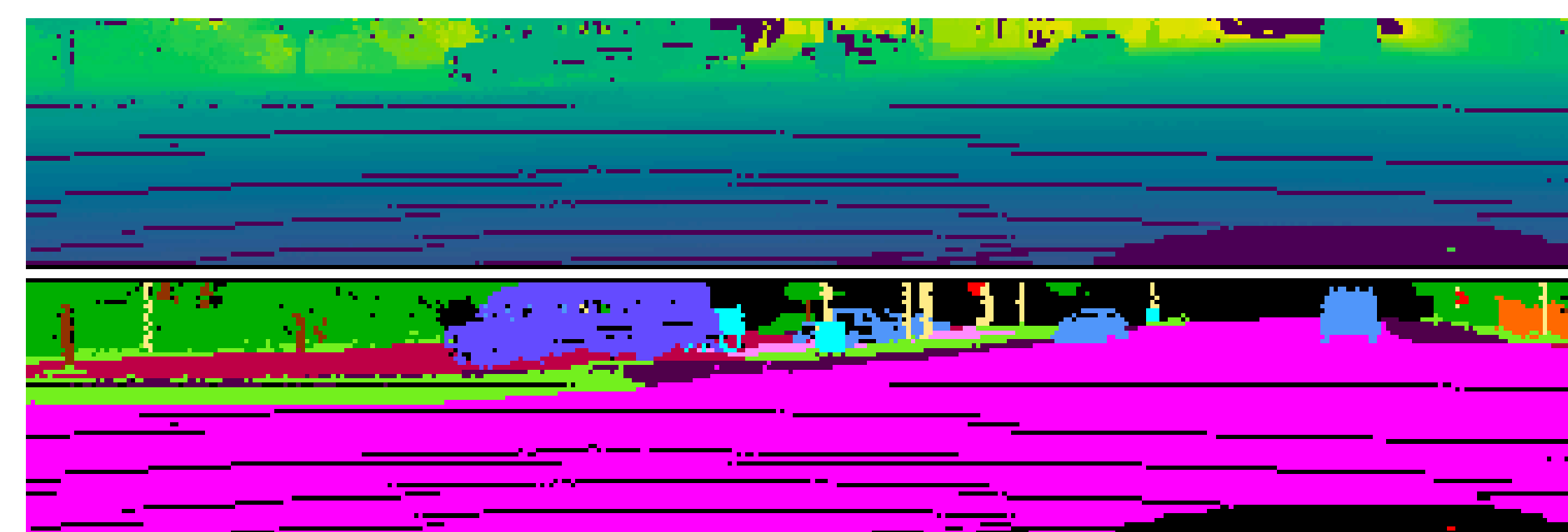
$w=512$: range image width

$$\phi = \sin^{-1}\left(\frac{z}{r}\right) \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$u = w \cdot \frac{1}{2} \left(1 - \frac{1}{\pi} \theta\right) \quad v = h \cdot \left(1 - \frac{\phi + f_{down}}{f}\right)$$



Ranges



Sources



Dataset

SemanticKITTI

The SemanticKITTI dataset is a large-scale benchmark of LiDAR scans captured by a Velodyne HDL-64E sensor. There are 22 sequences of driving data, comprising 43,000+ scans. Each frame contains approximately 100-130k points, with sequences 00-10 containing dense true labellings on 34 classes.

Evaluation Metrics

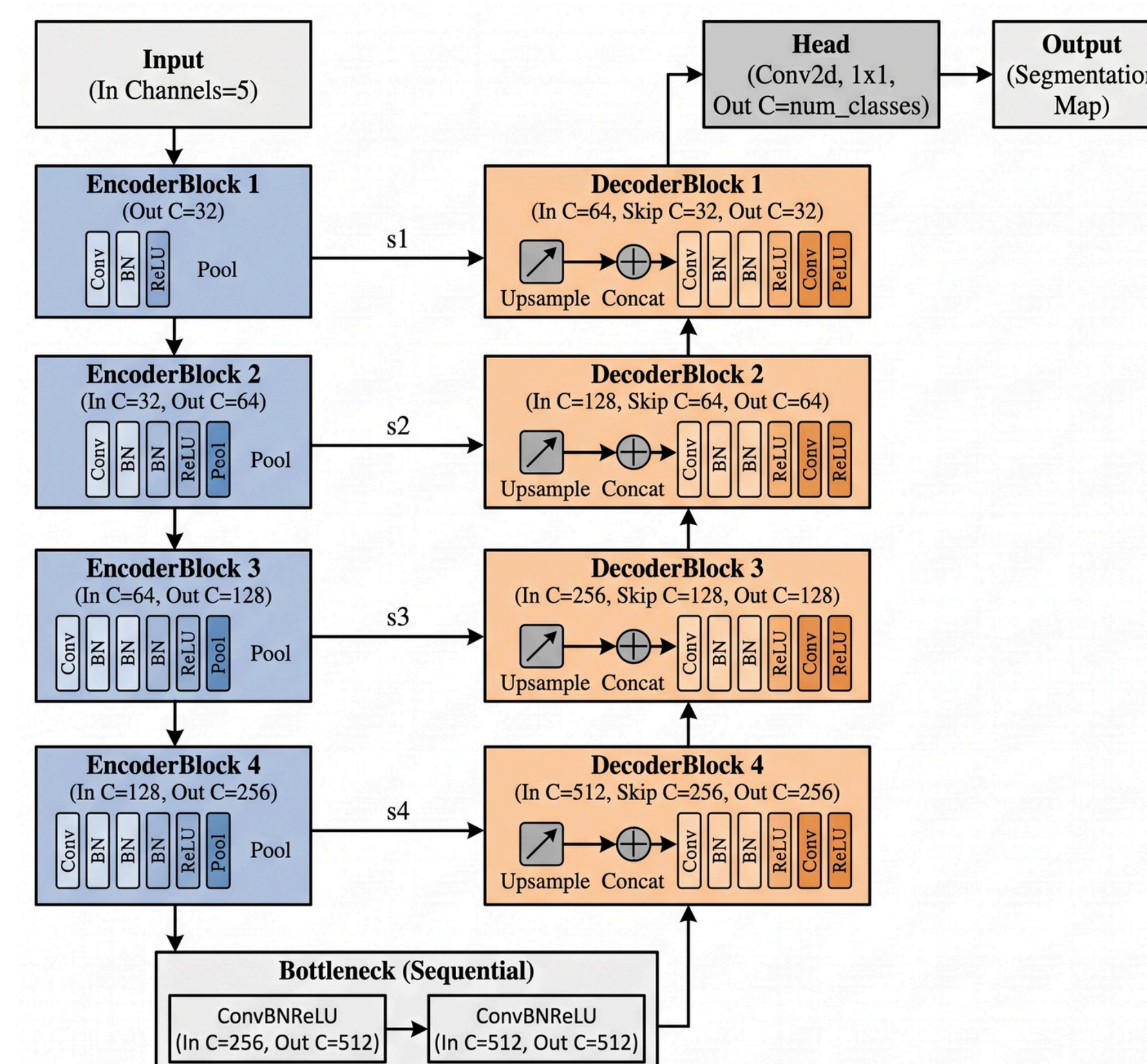
Intersection over Union (IoU) is computed for each class as the ratio of the number of correctly predicted points (intersection) to the total number of points that belong to a particular class in either the predictions or ground truth (union). An IoU of 1 indicates perfect agreement; 0 indicates no overlap.

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

CNN Architecture

Input: [5, H, W] tensor with ranges, xyz coordinates, and remission for nearest (u,v) pixels

Output: [H, W] tensor of predicted class labels



Post-Processing

window size=5x5; k=5

Gaussian STD=1.0

In projection-based segmentation, points that are stored in the same range image pixel will automatically have the same predicted class. To reduce resulting errors, we can average over k-nearest neighbors:

(1) Reproject predictions to 3D

Use stored pixel indices to map 2D range image predictions back to LiDAR points

(2) Define local neighborhoods

Gather nearby candidates within a 5x5 pixel window

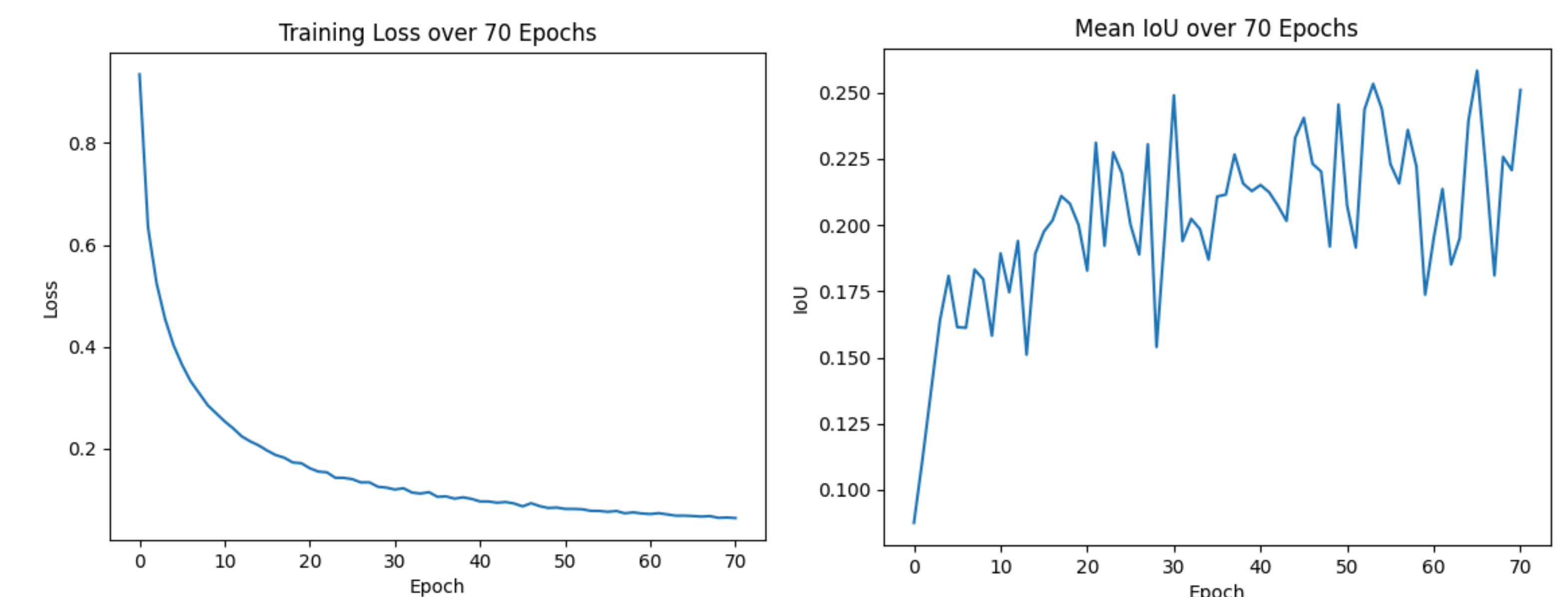
(3) Use ranges (Euclidean distances) to get relevant neighbors

(4) Apply Gaussian weighting

Penalize distant neighbors more strongly and give higher importance to closer ones

(5) Assign majority vote based on k closest neighbors

Results



Per-Object IoU

Mean IoU over selected classes: **0.39**

Ours
RangeNet

	Car	Bicycle	Motorcycle	Truck	Terrain	Person	Traffic-Sign	Road	Parking	Sidewalk	Building	Fence	Vegetation
Ours	0.4461	0.0685	0.0122	0.2410	0.5495	0.1914	0.0859	0.7910	0.2964	0.7561	0.7407	0.3947	0.4936
RangeNet	0.874	0.099	0.124	0.196	0.641	0.181	0.420	0.900	0.507	0.700	0.802	0.489	0.771

Discussion

Our model achieved a peak mIoU of approximately 25% over 70 epochs. While RangeNet++ reported 52.2% over 150 epochs, we trained on 5 sequences rather than the full 10 due to training time constraints. With roughly half the data, the model has less exposure to varied environments and rare classes like bicyclists and motorcyclists. Notably, the loss curve has not plateaued at epoch 70 and mIoU continues to climb, suggesting the model is still improving. Combined with our full implementation of the architecture and kNN postprocessing pipeline, this indicates that with access to the full dataset and additional training time, our implementation would very likely approach the results reported in the paper.